21.2.4 Irrigation

Water is important for the proper growth and development of plants. Plants absorb water from their surrounding with the help of the root system. The supply of water to crops at regular intervals is called **irrigation**. The time and frequency of irrigation varies from crop to crop, soil to soil and season to season. Fertilizers can also be applied through the irrigation. The various sources of irrigation are wells, tube wells, ponds, lakes, rivers, dams and canal. Effective irrigation is the controlled and uniform supply of water to crops, in the required amount at the right time with the minimum expenditure. Irrigation can be carried out by two different methods.

a. Traditional Methods
b. Modern Methods

a. Traditional Methods

In these methods, irrigation is done manually. Here, a farmer pulls out water from wells or canals by himself or using cattle and carries to farming fields. Pumps are also commonly used for lifting water from various sources. Diesel, biogas, electricity and solar energy are the few important sources of energy needed to run these pumps. The method of pulling water may vary from one place to other place. The main advantage of this method is that it is cheaper. But its efficiency is poor because of the uneven distribution of water. It also leads to heavy water loss.

Activity 3

Find out the irrigation system followed in your area. Also, debate on the advantages and disadvantages of modern irrigation systems like sprinkler system and drip system.

b. Modern Methods

The modern irrigation methods help to overcome the problems exist in the traditional methods. It also facilitates the even distribution of moisture in the field.

The modern methods involve two systems. They are:

- Sprinkler system
- Drip system

Sprinkler System

A sprinkler system sprinkles water over the crop and helps in an even distribution of water. This method is much advisable in areas facing water scarcity. Here a pump which generates pressure is connected to pipes, and water is sprinkled through the fine nozzles of pipes.



Figure 21.9 Sprinkler irrigation system

Drip System

In drip system, water is released drop by drop exactly at the root zone using a hose or pipe.



Figure 21.8 Traditional irritation methods

8th_Science_Unit-21_EM.indd 248 12/2/2022 8:05:03 PM